Project for Team C: Ship and Operate a Containerized App with a Full CI/CD Pipeline

Take a simple three-tier application (frontend, API, database) from source code to a live, monitored deployment, fully automated, with no manual deployment steps.

Use an existing open-source sample app rather than building one. The point is the DevOps work.. A small Node or Flask API with Postgres and a basic frontend.

Scope:

1. **Containerization:** Dockerfiles for each service, plus a docker-compose setup for local development.
2. **CI pipeline (GitHub Actions):** On every pull request, run linting, tests, and a security scan (Trivy for images, plus a dependency check). Build and push the image to a registry (Docker Hub or GitHub Container Registry).
3. **Infrastructure as Code (Terraform):** Provision the environment (a VM or small cluster on AWS free tier, or a local k3s/kind cluster if cost is a concern). State should be stored remotely.
4. **CD pipeline:** Merging to main automatically deploys to a staging environment. Promotion to production requires a manual approval step.
5. **Monitoring and alerting:** Prometheus and Grafana dashboards (CPU, memory, request rate, error rate), with one alert sent to Slack, Discord, or email when the app goes down.
6. **Documentation:** A README with an architecture diagram, setup steps, and a short runbook for "what to do if the deployment fails."

Suggested timeline (3 weeks)

Presentation outline (15 to 20 minutes)

1. The problem: why manual deployments are slow and risky (2 min)
2. Architecture overview (3 min)
3. **Live demo:** push a code change and watch it go through CI, deploy to staging, and appear in the dashboard (7 min)
4. **Failure demo:** break something on purpose and show the alert firing and the rollback (3 min)
5. Challenges faced and lessons learned (3 min)
6. Q&A